

Homework 5

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Model Specification and Fitting

1. Model formulation (10 points): Write out your model in the form $Y_i = X_i\beta + Z_ib_i + \epsilon_i$. Identify the fixed effects and the random effects. What assumptions are you making about the distributions of b_i and ϵ_i ?

Our linear mixed effects model is as follows:

$$Y_i = \beta_0 + \beta_1 x_{i,\text{YEAR}} + \beta_2 x_{i,\text{SEX}} + b_{i0} + b_{i1} x_{i,\text{YEAR}} + \epsilon_i$$

The fixed effect is $X_i\beta = \beta_0 + \beta_1 x_{i,\text{year}} + \beta_2 x_{i,\text{sex}}$. The random effect is $Z_ib_i = b_{i0} + b_{i1} x_{i,\text{year}}$ which represent a random intercept and a random slope for time. We assume that $b_i \sim N(0, D)$ and $\epsilon_i \sim N(0, R_i)$ where b_i independent of ϵ_i and D and R_i cannot both be unstructured covariance matrices.

We include a random intercept to allow each subject to have their own baseline SBP level which will capture individual variation in baseline SBPs after accounting for sex and time.

Our previous analyses have also shown that systolic blood pressure increases over time at the same rate for males and for females on average, but it is plausible that on an individual level, time still has a different effect on each subject's SBP levels, so our model includes a random slope for the number of years since baseline.

2. Random effects structure (15 points): Fit a linear mixed effects model to your data. Justify your choice of random effects structure (e.g., random intercepts only, or random intercepts and slopes). Report the fixed effect estimates and the estimated variance components.

The following is the summary of the linear mixed effects model that was specified in the previous section. We have a table of the model's fixed effects, followed by the estimated variance-covariance matrix for the random effects and the estimated marginal variance-covariance matrix.

Fixed effects summary

##	Value	Std.Error	DF	t-value	p-value
## (Intercept)	129.2361382	0.52416974	6411	246.553984	0.000000e+00
## YEAR	0.8379731	0.02790807	6411	30.026196	1.888545e-185
## SEX	1.8536034	0.67245289	3204	2.756481	5.875789e-03

Random effects variance covariance matrix

##	(Intercept)	YEAR
## (Intercept)	293.900000	-0.099488
## YEAR	-0.099488	0.699060
## Standard Deviations:	17.144	0.8361

```
## RANDID 6238
## Conditional variance covariance matrix
##      1      2      3
## 1 129.45    0.00    0.00
## 2    0.00 129.45    0.00
## 3    0.00    0.00 129.45
## Standard Deviations: 11.378 11.378 11.378
```

Evaluating the Random Effects

1. Testing random effects (15 points): Evaluate whether the random effects in your model are needed. Describe your approach and interpret your findings. If your test involves a boundary issue, discuss how you addressed it.

To test whether or not the random slope effect for time is needed, we fit a model with only a random intercept and use its log likelihood, along with the log likelihood of our original model, to perform a likelihood ratio test statistic.

We are testing whether or not the variance of the random effect for time is zero, which is on the boundary of the parameter space. To correct for this, we will compare our test statistic to a mixture distribution. Since the new model has one random effect and the original model has two random effects, we will compare the test statistic to $\frac{1}{2}\chi_1^2 + \frac{1}{2}\chi_2^2$.

```
## Likelihood Ratio Test:
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```
##      chi.2 p
## 1 146.2361 0
```

The LRT test stat is 146.23 and yields a p -value less than 0.05, so we reject the null hypothesis and conclude that the variance of the random slope effect for time is not zero. Therefore, the random effect should remain in the model.

2. Variance components interpretation (10 points): Interpret the estimated variance components. What do they tell you about the sources of variability in your data (between-subject vs. within-subject)?

The estimated variance of the random intercepts is 293.9, meaning that on an individual level, the standard deviation of the deviations from the estimated fixed baseline SBP measure, after controlling for the fixed effects of sex and time, is $\sqrt{293.9} = 17.144$.

The estimated variance of the random slopes for the time effect is 0.699, meaning that on an individual level, the standard deviation of the deviations from the estimated fixed effect of time on SBP, after accounting for the fixed baseline SBP and the fixed effect of sex, is $\sqrt{0.699} = 0.836$.

The estimated covariance between the random intercepts and the random slopes is -0.099 , meaning that on average, as an individual's random intercept increases, the random slope for the time effect slightly decreases.

The estimated conditional variance of SBP at each time point is 129.45, meaning that after random effects are accounted for, the standard deviation of SBP levels at each fixed time point is $\sqrt{129.45} = 11.378$.

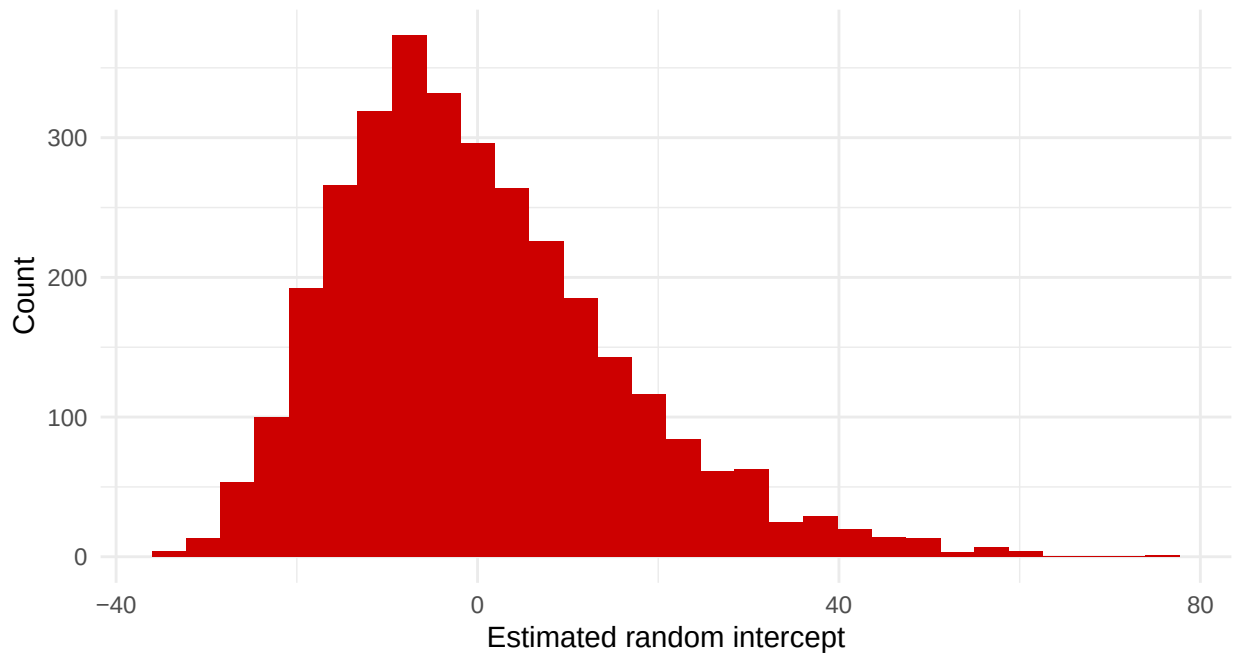
Since the variance of the random intercepts is greater than the conditional variance of SBP, we know that the overall variability in SBP is due more to between-subject variation and due less to within-subject variation.

Subject-Specific Predictions

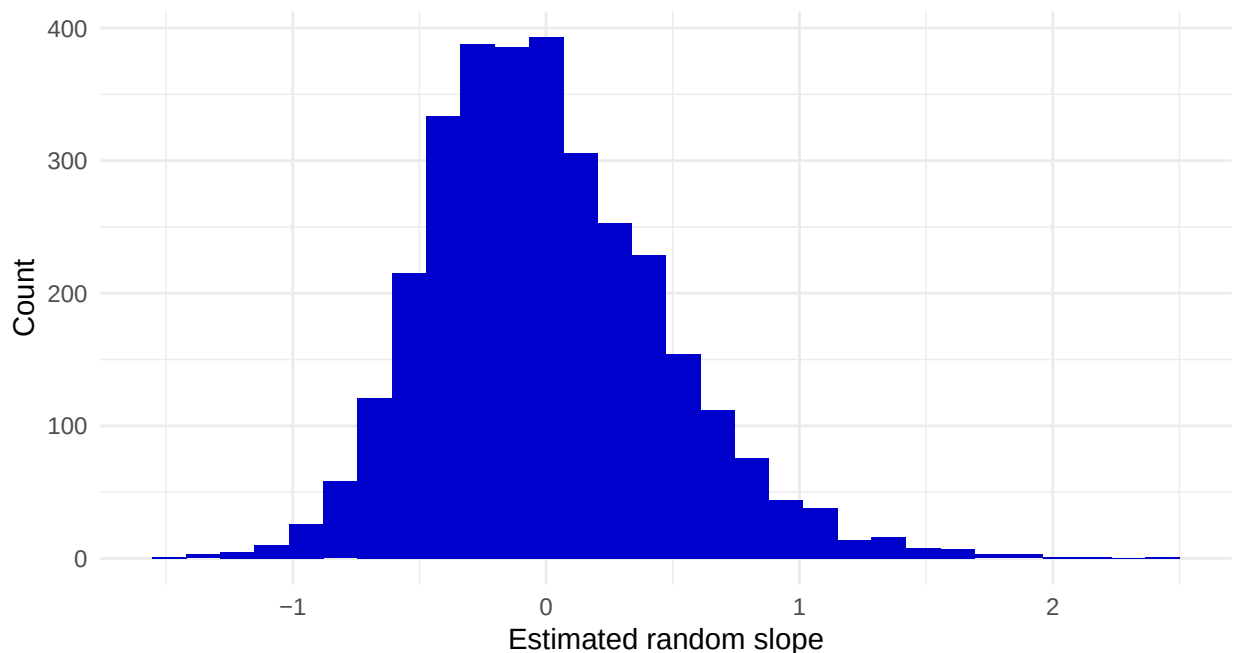
1. **BLUPs (10 points):** Obtain the predicted random effects (empirical BLUPs) for each subject. Create a figure displaying these predictions (e.g., a caterpillar plot or histogram of random intercepts/slopes).

We use the ‘ranef’ function to compute the predicted random effects for the model which include the random intercept and the random slope for year. We plot their empirical distributions below.

Empirical BLUPs: Random Intercepts

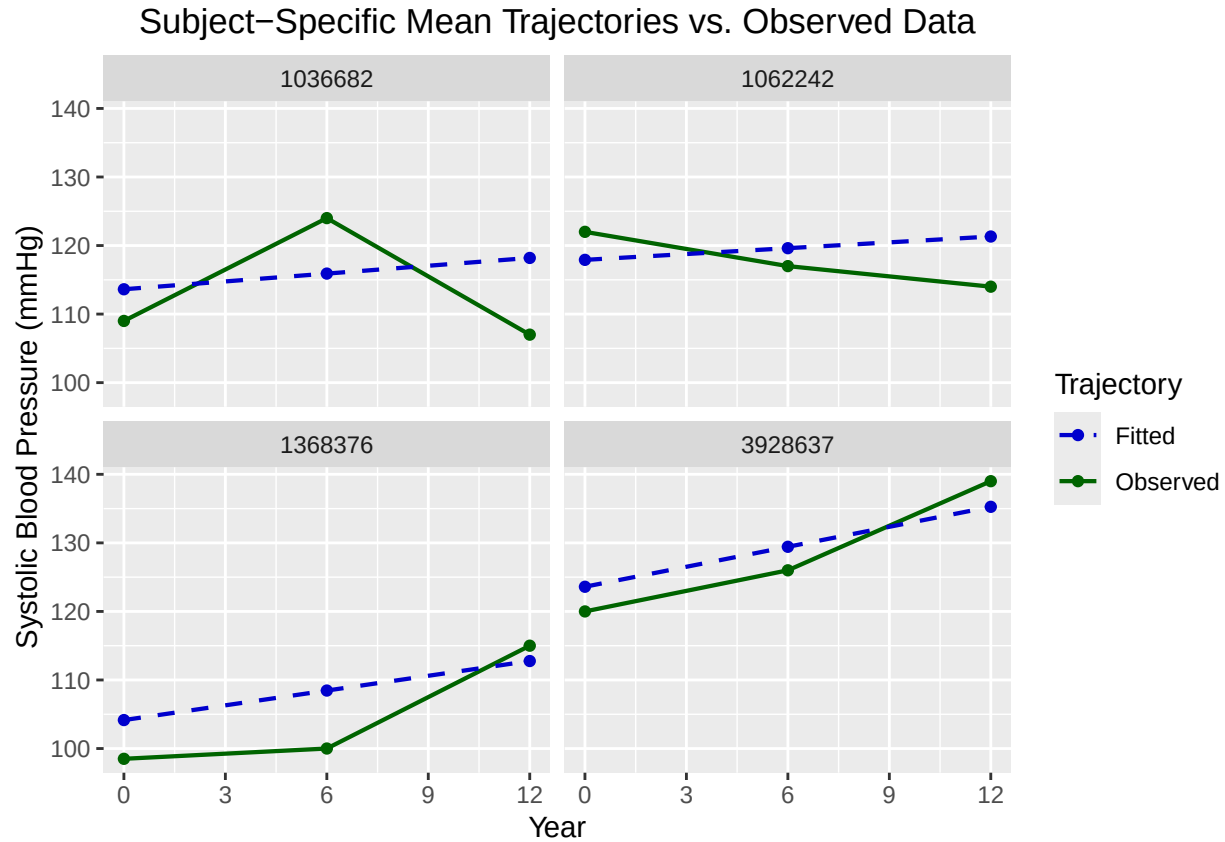


Empirical BLUPs: Random Slopes for YEAR



2. Subject-specific trajectories (10 points): For a subset of subjects, plot the observed data along with the subject-specific fitted trajectories. Comment on how well the model captures individual patterns.

For four randomly selected patients, we plot their observed data with a solid green line and their subject-specific mean trajectory with a blue dotted line.



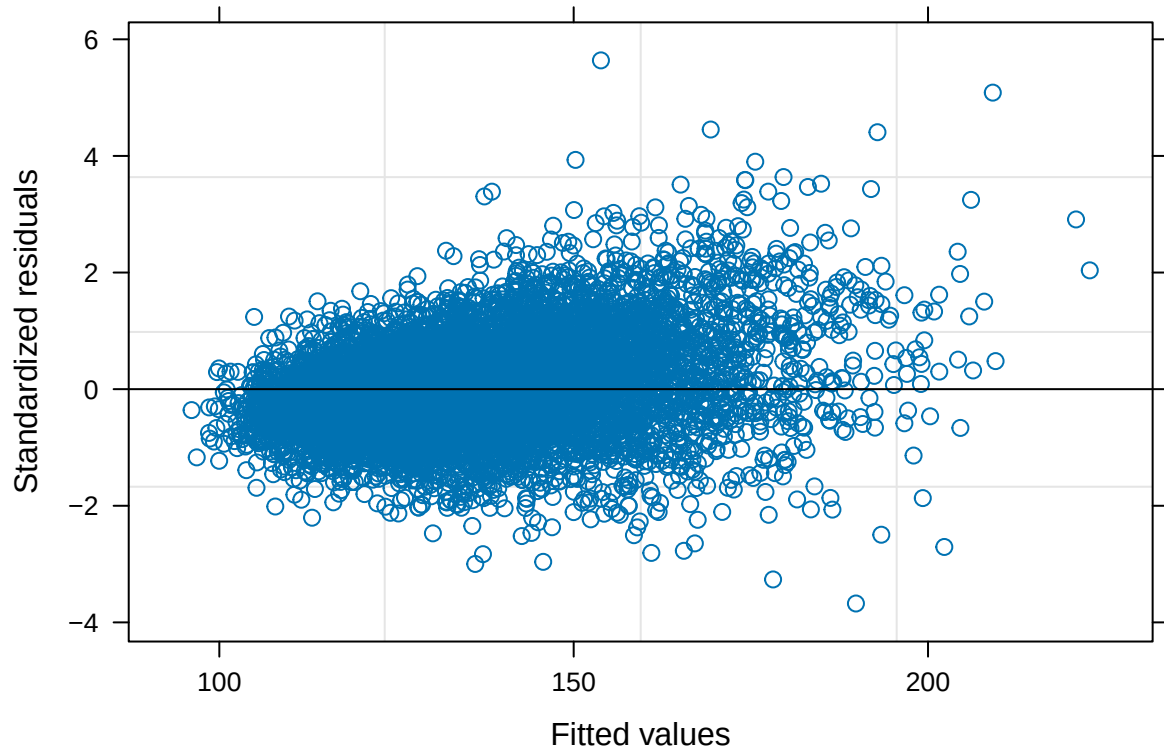
For two of the patients, the upward trend of the SSMT matches the general upward trend of the observed data. For the other two patients, however, the SSMT is a poor fit, since the trajectory of the observed data for the bottom left patient is nonlinear, and the trajectory of the observed data for the bottom right patient is a downward trend.

Thus we see that the model does not always capture individual trends well, but this is to be expected in the context of our analysis, where thousands of patients were tracked over twelve years, which leaves plenty of room for individual deviations from the average trends in SBP over time.

Residual and Influence Diagnostics

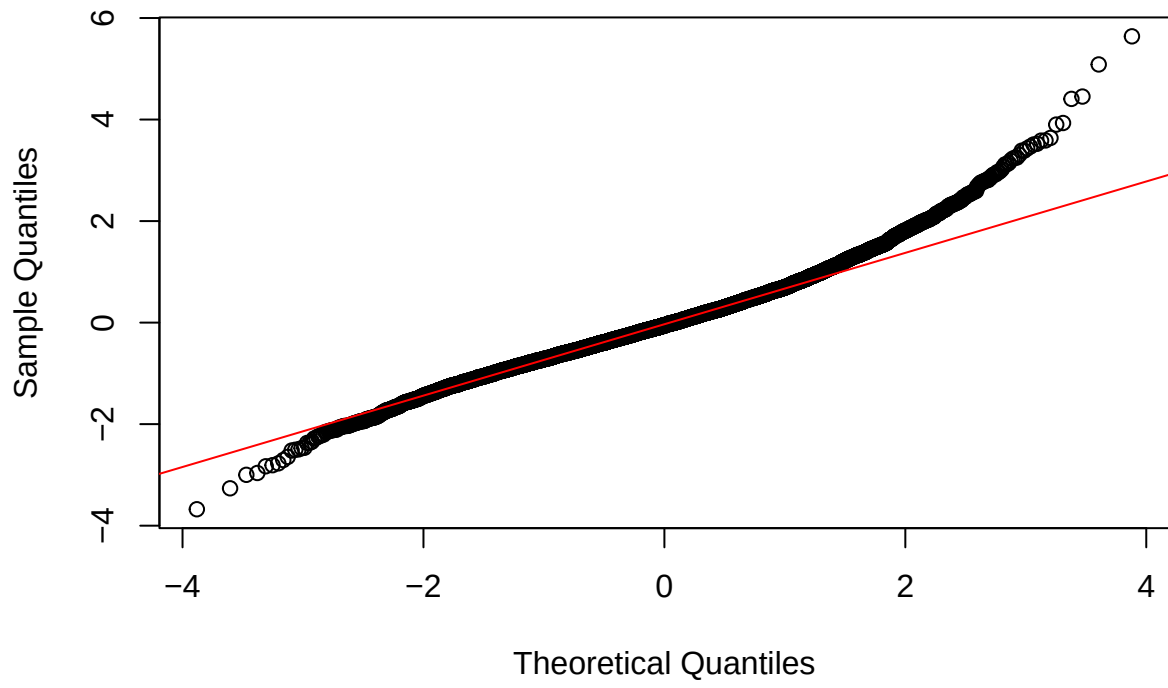
1. Residual analysis (10 points): Examine the residuals from your model. Create appropriate diagnostic plots and comment on whether the model assumptions appear to be met.

To examine the residuals we check a few different plots. First, we can look at the residuals versus fitted plot. Ideally, we want this plot to be centered around 0 and to show no patterns. Looking at the plot, it does seem to be centered around 0, however as the fitted values increase, there is more of a deviation away from the 0 line. This suggests potential heteroskedasticity.

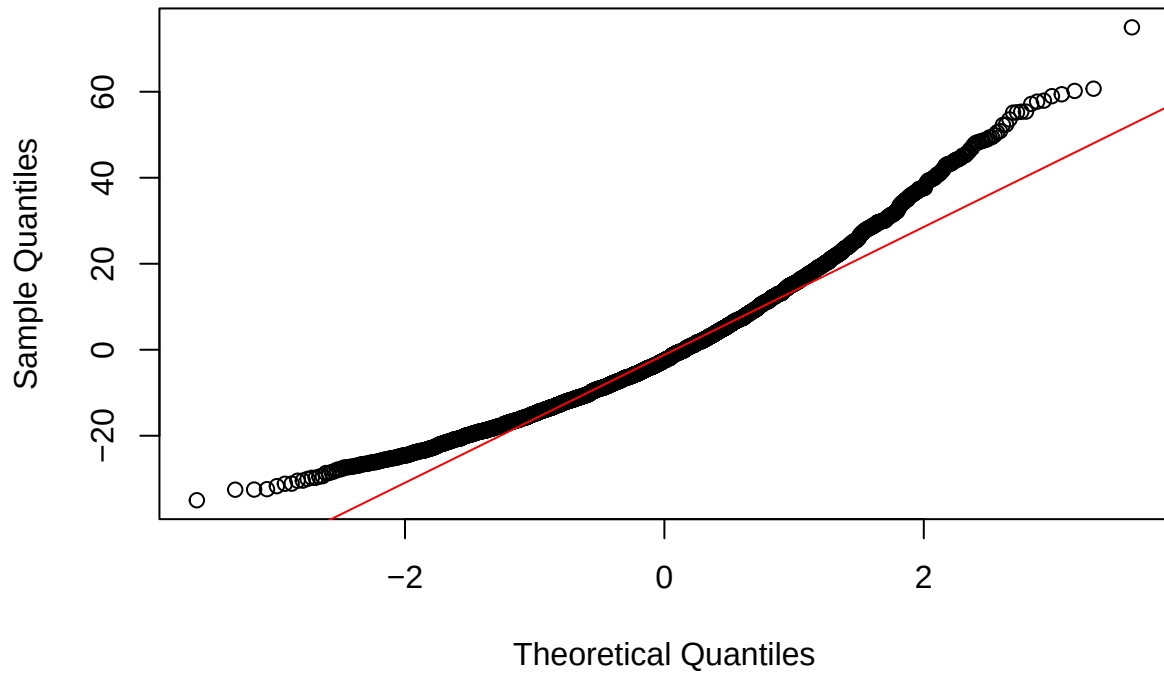


Next, we can look at the QQ plots for the residuals, random intercepts, and the random slopes. We want these plots to look approximately linear with minimal tail deviations. Looking at all three plots below, we see larger deviations on the tails than we would like. This suggests that the assumption of normality for both the errors and random effects may be violated.

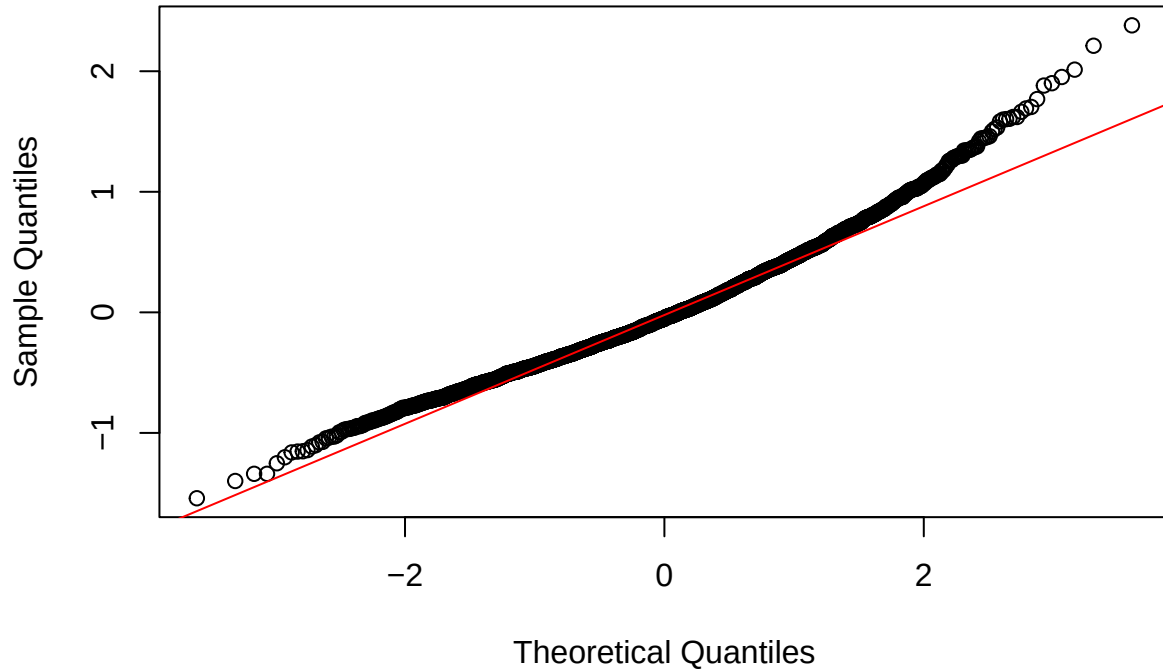
Normal Q-Q Plot



QQ Plot: Random Intercepts



QQ Plot: Random Slopes



2. Influence diagnostics (10 points): Conduct influence diagnostics to identify any subjects with unusual influence on the model fit. Describe what you found and discuss whether any observations warrant further investigation.

In order to test the influence, we will first find the observations that have the most extreme random intercepts and random slopes. Then, one at a time, we will refit the model on our sample that leaves one of these extreme points out. We compare the original effects to the effects of each of these models to see if there are any drastic changes. The table below depicts the absolute difference in the estimates from the original model and the leave-one-out models. Overall, the differences are quite small numbers. We find that the subject 5491585 has the greatest influence on the intercept and the sex coefficient, however these changes are relatively small in magnitude indicating that this subject does not meaningfully alter the conclusions of the analysis.

##	ID	Intercept	Year Coefficient	Sex Coefficeint
## 1	2754554	0.010334513	0.0021205324	0.047071015
## 2	9837913	0.001261062	0.0002614581	0.039559477
## 3	5491585	0.052139911	0.0002585627	0.053180260
## 4	2290290	0.006356269	0.0013275007	0.037842065
## 5	6193708	0.004123336	0.0008594820	0.037927053
## 6	5807368	0.010420630	0.0022086407	0.027402305
## 7	2380239	0.008943690	0.0018446262	0.037861809
## 8	7979046	0.010242680	0.0020006324	0.018754165
## 9	4339221	0.007082000	0.0019746314	0.015518510
## 10	4441180	0.008910260	0.0020006324	0.007220942

Summary and Interpretation

Summarize your findings as you would for a collaborator. Address: - What is your fitted model and what random effects structure did you choose? - What are the key findings regarding your research question? - Are there any concerns about model fit or influential observations?

After a thorough investigation of the data, the fitted model we chose is a linear mixed effects model that includes both a random intercept and a random slope for year. We chose a model with these random effects in order to account for individual variation of both baseline SBP and the effect of time on SBP. The likelihood ratio test confirmed that the random slope for time was necessary. We specify the model as follows:

$$Y_i = \beta_0 + \beta_1 x_{i,\text{YEAR}} + \beta_2 x_{i,\text{SEX}} + b_{i0} + b_{i1} x_{i,\text{YEAR}} + \epsilon_i$$

The fitted linear mixed effects model indicates that SBP increases over time on average and that there is a difference in the SBP between males and females. The relatively large variance of the random intercepts compared to the residual variance suggests that between subject variability in the baseline SBP is a major source of variation in the data.

The diagnostic plots suggest that there are some departures from the model's assumptions, namely possible heteroskedasticity and deviations from the normality assumptions for both the residuals and the random effects. These patterns suggest that the model may not be fully capturing the underlying data structure.

Lastly, we found a handful of data points that could be considered influential observations, but after leave-one-out analyses, we found that these points led to small changes in the model's estimates. This suggests that no single subject has a strong influence on the model.